

Decentralized Identity and Access Management (IAM) using Blockchain

Mr.R.Raja Sekar

*Assistant professor, computer science
and engineering
Kalasalingam academy of research
and education
Krishnankovil, TamilNadu
rsblacktulip@gmail.com*

Aman Abraham

*Computer science and engineering
Kalasalingam academy of research
and education
Krishnankovil, TamilNadu
amanabrahamx@gmail.com*

Abhiram Masna

*Computer science and engineering
Kalasalingam academy of research
and education
Krishnankovil, TamilNadu
masnaabhiram2002@gmail.com*

Pavan Reddy Pagilla

*Computer science and engineering
Kalasalingam academy of research
and education
Krishnankovil, TamilNadu
pagillapavan161@gmail.com*

Sagar Sharma

*Computer science and engineering
Kalasalingam academy of research
and education
Krishnankovil, TamilNadu
sagar1012002sharma@gmail.com*

Abstract—In this transformative initiative, Identity and Access Management (IAM) undergoes a profound evolution through decentralized blockchain technology. The project focuses on developing a user-centric identity wallet, leveraging blockchain's attributes for transparency, immutability, and cryptographic security, orchestrating smart contracts to automate identity verification and access control. This decentralized IAM architecture not only fortifies security but also empowers users with ownership and sovereignty over their personal data, ensuring a tamper-resistant environment and fostering a more resilient and user-centric digital landscape. The project's strategic alignment with interoperability standards amplifies its impact, promising to set new benchmarks for secure and efficient IAM frameworks across industries.

Keywords—Decentralized Identity, IAM: Identity and Access Management, Blockchain, Smart Contracts, Verifiable Credentials, Security, Privacy, User-Centric, Interoperability.

I. INTRODUCTION

In navigating the ever-evolving landscape of digital transactions and interactions, the imperative to redefine Identity and Access Management (IAM) systems becomes increasingly pronounced. The traditional paradigms of IAM, characterized by centralized databases fraught with vulnerabilities such as single points of failure and security breaches, underscore the pressing need for a revolutionary reimagining. In response to these multifaceted challenges, our project undertakes a groundbreaking initiative, harnessing the transformative potential encapsulated in blockchain technology. This endeavor is not merely a technical upgrade but a holistic response, aimed at recalibrating the fundamental dynamics of IAM.

At its essence, our project aspires to birth a decentralized IAM system that transcends the conventional boundaries of security and privacy. Central to this ambition is the pivotal shift in control, placing the custodianship of personal data

squarely in the hands of the individual, thus empowering them in unprecedented ways. The trajectory of this endeavor begins with meticulous research, where we delve into the nuanced selection of blockchain platforms and consensus mechanisms. This careful curation is paramount in aligning with the robust requirements intrinsic to a decentralized IAM architecture.

Moving into the design and development phase, our focus intensifies on the creation of a user-friendly decentralized identity wallet—a digital fortress where users can wield unprecedented control over their identity information. This initiative is complemented by the integration of smart contracts, intricate self-executing codes embedded within the blockchain, serving as the linchpin for seamless identity verification, authentication, and access control. Diving deeper into the technological intricacies, the incorporation of decentralized identifiers (DIDs) and verifiable credentials emerges as a testament to our commitment to constructing a secure, privacy-preserving environment. These elements act as formidable safeguards, substantially mitigating the risks associated with unauthorized access and breaches. Beyond this, the establishment of interoperability standards is pivotal, ensuring that our decentralized IAM system harmoniously integrates with the existing fabric of applications and services, fostering a bridge between innovation and compatibility.

As the project progresses, the scrutiny intensifies through rigorous testing procedures. These evaluations are designed to subject the system to exhaustive examinations, meticulously dissecting its security architecture and operational reliability. The overarching goal is not merely to meet industry standards but to surpass them, setting new benchmarks in the realms of security, privacy, and operational efficiency. In conclusion, this project encapsulates more than a mere technological advancement;

it symbolizes a paradigmatic shift in how we conceptualize, implement, and safeguard digital identities.

By boldly addressing the intricate complexities entrenched in traditional IAM systems and ushering in a decentralized alternative, our endeavor aspires not just to mitigate vulnerabilities but to pioneer a new era where security, privacy, and efficiency converge seamlessly in the digital sphere.

II. RELATED WORKS

II.1 *Background Study*

The landscape of decentralized Identity and Access Management (IAM) has undergone a transformative shift in recent years, propelled by advancements in blockchain technology. This literature survey navigates through the extensive body of research and scholarly discourse surrounding the intersection of decentralized IAM and blockchain. In a digital ecosystem rife with challenges associated with centralized identity systems, researchers and practitioners have increasingly turned their attention to the promises and pitfalls of decentralized solutions. The survey encapsulates a comprehensive analysis of existing literature, spanning seminal works, contemporary research papers, and theoretical frameworks. By synthesizing the collective knowledge in this domain, the survey aims to elucidate the key trends, theoretical underpinnings, and methodological approaches that have shaped the discourse on decentralized IAM. Through an exploration of blockchain's role in fortifying security, empowering user control, and addressing the vulnerabilities of traditional IAM systems, this survey endeavors to provide a nuanced understanding of the current state of knowledge in this evolving field.

This study talks about how Cloud computing's expansion has brought economic benefits, but its centralized trust models face security challenges, prompting the exploration of blockchain's decentralized nature for building trust infrastructure. This study delves into migrating data and permissions from centralized cloud systems to decentralized file systems like IPFS, emphasizing content-addressed storage without privileged nodes [1].

This paper highlights the rising need for digital identities in interconnected smart vehicle systems and proposes a solution utilizing Self-Sovereign Identities and blockchain to decentralize identity management. By examining use cases and interactions between involved entities like vehicle owners, drivers, and service providers, it demonstrates how decentralized identity management ensures authentication and authorization processes. The system's feasibility in real-life scenarios is substantiated through testing and measurements, emphasizing its practical application [2].

This paper outlines the user benefits and ownership advantages from decentralized digital identity management while summarizing the current status and regulatory aspects of Identity and Access Management within blockchain environments, proposing solutions based on existing literature and past experiences [3].

This paper explores the impact of blockchain technology on cybersecurity, emphasizing the need for stronger digital identity protection in online interactions. It introduces 'MedSecureChain,' a blockchain-based identity and access management system implemented within a medical ecosystem. Utilizing OAuth-based authentication and smart contracts, the system ensures user data control, document verification through IPFS and blockchain, enhancing security, privacy, and decentralization in safeguarding user identities [4].

This research paper emphasizes the limitations of current Cloud Identity and Access Management solutions, citing their inefficiency, high costs, and susceptibility to data breaches. It advocates for a new approach utilizing blockchain technology, focusing on its decentralized nature, security features, and potential to transform identity management processes. The chapter provides a thorough analysis of blockchain's characteristics, highlighting its relevance and future enhancements, while pinpointing significant research gaps and directions for further exploration in the field of identity and access management [5].

This research delves into the evolving landscape of identity management, highlighting the shift towards decentralized solutions using blockchain technology, empowering users with complete control over their data. It explores the significance of Self-Sovereign Identity (SSI) management facilitated by blockchain, addressing its potential to resolve privacy concerns and scalability issues while investigating the utility of blockchain in Identity and Access Management (IAM) with SSI through an exploration of key enabling technologies [6].

This paper advocates for a decentralized IAM solution, addressing the privacy concerns associated with centralized systems by utilizing blockchain as a secure and decentralized backbone. In this proposed solution, sensitive user identity information remains stored on the user's mobile device, ensuring both privacy protection and user control over their identity data [7].

This paper examines the evolution of blockchain-based identity solutions, comparing them with established systems and conducting a comprehensive survey across various criteria, highlighting both strengths and weaknesses. By analyzing requirements encompassing ecosystem factors, user functionality, compliance, regulations, and integration, the paper provides insights into a diverse range of solutions within this expanding field [8].

III. INVESTIGATIONS OF CURRENT PROJECT AND RELATED WORK

III.1 *Existing Approach*

In examining the existing approaches to Identity and Access Management (IAM), it is evident that conventional systems predominantly rely on centralized architectures. Traditional IAM solutions typically involve the creation of centralized databases where user identity information is stored and managed by a single authoritative entity, such as an organization or a service provider. In this paradigm, users

rely on a username and password combination for authentication, with access permissions determined and controlled centrally.

While these conventional IAM systems have been widely adopted and are well-established, they exhibit inherent vulnerabilities that have become increasingly apparent in the face of evolving cybersecurity threats and privacy concerns. Centralized databases represent single points of failure, making them attractive targets for malicious actors seeking unauthorized access to a trove of sensitive user information. Furthermore, the reliance on usernames and passwords is susceptible to various security risks, including phishing attacks, credential theft, and password breaches.

In response to these shortcomings, there has been a growing interest in decentralized approaches to IAM. Blockchain technology, in particular, has emerged as a transformative force in redefining IAM paradigms. The decentralized IAM systems leverage blockchain's core features, such as decentralization, cryptographic security, and transparency, to create a more robust and user-centric model. These systems distribute identity information across a network of nodes, eliminating single points of failure and providing users with greater control over their own data through cryptographic keys.

III.II Problem Definition

The exploration of decentralized Identity and Access Management (IAM) within the context of blockchain technology necessitates a rigorous problem definition that encapsulates the nuanced challenges and opportunities inherent in this evolving domain. As illuminated by the literature survey, the integration of blockchain in IAM introduces complexities such as scalability concerns and interoperability challenges that demand meticulous attention. Additionally, the impact of decentralized IAM on regulatory compliance and legal frameworks emerges as a pivotal area of inquiry, reflecting the need for a sophisticated understanding of the intersection between technological innovation and legal parameters. While the theoretical promises of heightened security, user autonomy, and the mitigation of vulnerabilities in traditional IAM systems are discernible, practical implementations encounter real-world intricacies that merit meticulous examination. This refined problem definition serves as a cornerstone, guiding scholarly inquiry towards addressing these complexities and advancing the field with a comprehensive awareness of both theoretical promises and practical challenges.

III.III Objective And Scope Of The Project

The primary objective of this project is to develop a decentralized Identity and Access Management (IAM) system, leveraging blockchain technology to significantly enhance security, privacy, and user control over personal data.

1. Enhance Security, Privacy, and User Control: Develop a decentralized IAM system using blockchain to

fortify security measures, safeguard user privacy, and empower individuals with increased control over their identity information.

2. Streamline Identity Verification and Access Management: Implement smart contracts to automate and streamline identity verification and access management processes, reducing administrative burdens and enhancing user-friendliness.

3. Foster Transparency and Trust: Create a transparent and auditable identity management system through the immutability of blockchain, fostering trust between users and service providers by allowing independent verification of identity-related activities.

4. Establish Interoperability Standards: Ensure seamless integration of the decentralized IAM system with existing applications and services by adhering to widely accepted protocols and APIs, facilitating a frictionless user experience across diverse platforms.

The scope of this project is comprehensive, encompassing various key aspects of decentralized IAM implementation. It commences with meticulous research to identify and select the most appropriate blockchain platforms and consensus mechanisms tailored to the unique requirements of decentralized IAM. The subsequent design and development phase focuses on crafting a user-friendly decentralized identity wallet, providing users with a secure space to manage and control their identity information seamlessly. The implementation of smart contracts will streamline identity verification, authentication, and access control processes, contributing to the overall efficiency of the system.

Exploration of decentralized identifiers (DIDs) and verifiable credentials represents a crucial facet of the project, emphasizing the adoption of secure and privacy-preserving methods for identity verification. Integration with existing applications and services is a paramount goal, necessitating adherence to standard protocols to ensure compatibility and interoperability. Rigorous testing procedures will be employed to validate the security and reliability of the decentralized IAM system, ensuring its robustness in real-world scenarios. In essence, the project's scope spans from foundational research and platform selection to the development of user-centric applications, implementation of cutting-edge technologies, and rigorous testing, all with the overarching goal of revolutionizing IAM in a decentralized paradigm.

III.IV Solution

The solution to the critical challenges in Identity and Access Management (IAM) lies in the strategic implementation of a decentralized IAM system, harnessing the transformative potential of blockchain technology. In response to the pressing need for heightened security and a more user-centric approach in today's digital landscape, this project redefines the conventional IAM framework. The deployment of blockchain's cryptographic principles ensures an unparalleled level of security, mitigating risks associated

with data breaches and identity theft. Crucially, users are empowered with ownership and control over their identity information, a stark departure from traditional systems where centralized entities hold sway. This empowerment is fortified by the unique cryptographic keys wielded by users, ensuring that access to personal data is contingent on their explicit consent.

The integration of smart contracts within the blockchain framework propels efficiency in identity verification and access management. Automation of routine tasks not only streamlines processes but also enhances the user experience, making interactions secure and seamless. Transparency and immutability, inherent features of blockchain, create a tamper-resistant system, instilling trust by allowing users and service providers to audit and verify identity transactions. The decentralized nature of the solution, by eliminating single points of failure, offers a robust defense against security breaches.

Moreover, the project prioritizes interoperability standards, ensuring the seamless integration of the decentralized IAM system with existing applications and services. By adhering to widely accepted protocols and open standards, the system guarantees a user-friendly experience, enabling individuals to effortlessly use their decentralized identities across diverse platforms. In essence, the solution doesn't merely address the identified challenges but sets forth a transformative vision for IAM, one where security, privacy, and user control converge harmoniously to meet the evolving demands of our digital era.

IV. METHODOLOGY

IV.I Components

- **Smart Contracts Development:** Detail the process of designing and coding smart contracts for identity management, access control, and auditing. Describe how these contracts are deployed on the chosen blockchain platform.
- **User Interfaces Implementation:** Explain the development of user interfaces, whether web or mobile applications, and how they interact with the blockchain. Detail the user interactions for functions such as identity management and resource access.
- **Identity Creation and Registration Module:** Outline the coding and implementation of the module that allows users to create blockchain-based identities and register real-world attributes securely.
- **Authentication Module:** Describe the implementation of authentication methods, such as digital signatures or zero-knowledge proofs, ensuring secure user access.
- **Authorization Module:** Explain the logic and coding behind the access control module within

smart contracts to enforce policies and regulate resource access.

- **Privacy-Enhancing Features:** Detail the integration of privacy-centric features like zero-knowledge proofs to safeguard user data during identity verification.
- **Security Measures Implementation:** Explain the coding and implementation of comprehensive security measures to fortify user data protection and ensure the integrity of the blockchain.

IV.II Model Description

IV.II.I Functional Interactions:

In the proposed system, users engage with the Identity Creation and Registration Module to establish blockchain-based identities using the provided User Interfaces. Subsequent user access attempts trigger the Authentication Module, which employs cryptographic methods to verify user identities. The Authorization Module, embedded within smart contracts, then governs access permissions by evaluating the authenticated user's identity against predefined rules. Importantly, smart contracts meticulously maintain an immutable record of identity-related activities, ensuring a transparent and auditable trail for monitoring and accountability. This seamless process allows users to interact effortlessly with the system through intuitive User Interfaces, enabling them to manage their identities and securely access resources.

IV.II.II Model Flow:

User Interaction: Users initiate interactions through the User Interfaces, performing actions like identity creation, access requests, and resource management.

Blockchain Execution: User requests trigger the execution of smart contracts on the blockchain, enforcing identity and access management logic.

Identity Verification: During access attempts, the Authentication Module verifies user identities, ensuring secure and authenticated interactions.

Access Control: The Authorization Module regulates access permissions based on the authenticated user's identity and predefined policies.

Logging and Auditing: Smart contracts log identity-related activities, providing an immutable record for auditing and accountability.

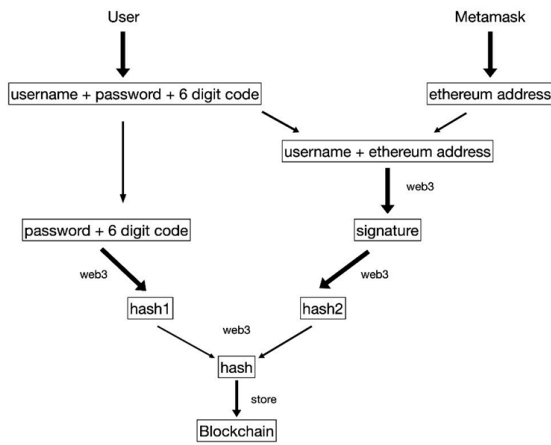


Fig 1. Flow Chart

IV.III. Model Architecture

The decentralized Identity and Access Management (IAM) system using blockchain can be delineated in a stepwise model architecture, encompassing the following key components and interactions:

- **Smart Contracts Layer:** These contracts execute predefined rules on the blockchain, ensuring secure and tamper-resistant operations.
- **User Interfaces Layer:** Users initiate actions, such as identity creation, access requests, and resource management.
- **Identity Modules Layer:** Interacts with the Smart Contracts Layer to create blockchain-based identities. Verification is conducted through the Authentication Module, which may employ privacy-enhancing features like zero-knowledge proofs.
- **Access Control Layer:** Interacts with the Access Control Smart Contract in the Smart Contracts Layer.
- **Privacy-Enhancing Features Layer:** Integrated with the Authentication Module for enhanced privacy during identity verification.
- **Security Measures Layer:** Encryption mechanisms secure user data within the system. Blockchain integrity checks ensure the trustworthiness of the distributed ledger.
- **Model Flow:** User Interaction: Users interact with the User Interfaces Layer to initiate identity-related actions. Identity Verification: Identity Creation and Registration Module interacts with the Smart Contracts Layer for identity creation. Users' identities are verified through the Authentication Module, potentially using privacy-enhancing features. Access Control and Auditing: The Access Control Layer enforces access policies based on authenticated identities. Auditing Smart Contract logs and stores identity-related activities for

transparency and accountability. Blockchain Execution: The Smart Contracts Layer executes identity management and access control logic on the blockchain, ensuring decentralized and tamper-resistant operations. This step-wise model architecture provides a structured overview of the decentralized IAM system, delineating the progression of actions and interactions within each layer.

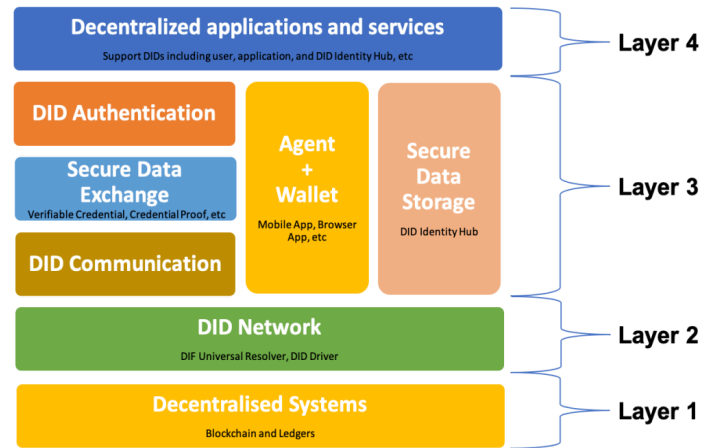


Fig 2. Decentralized application model

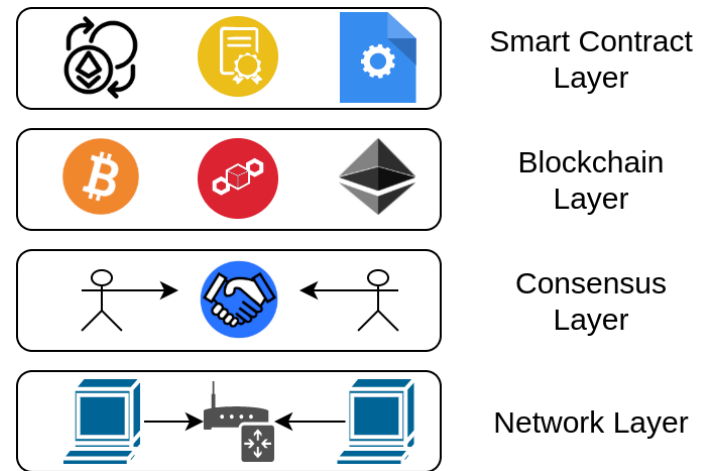


Fig 3. Layers of the blockchain

IV.IV. Methodology Adopted

The methodology employed for this project encompasses a systematic approach to implementing decentralized Identity and Access Management (IAM) using blockchain technology. The first phase involves defining clear objectives, outlining the goals of the decentralized IAM system, and identifying specific requirements, including user authentication, authorization, and digital identity management. Following this, a careful selection of a suitable blockchain platform is made, considering factors such as scalability, consensus mechanisms, and smart

contract capabilities. The data model for storing decentralized identities on the blockchain is meticulously designed, specifying attributes and information associated with each user.

The project progresses with the development of smart contracts to manage identity and access rules. These smart contracts include functions for user registration, authentication, authorization, and attribute management. The implementation of a mechanism for users to create decentralized identities on the blockchain follows, leveraging public-private key pairs for secure identity creation and management. The security of the authentication process is ensured through the use of cryptographic techniques to maintain the integrity and confidentiality of authentication processes.

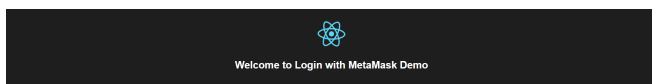


Fig 4. Sign In page

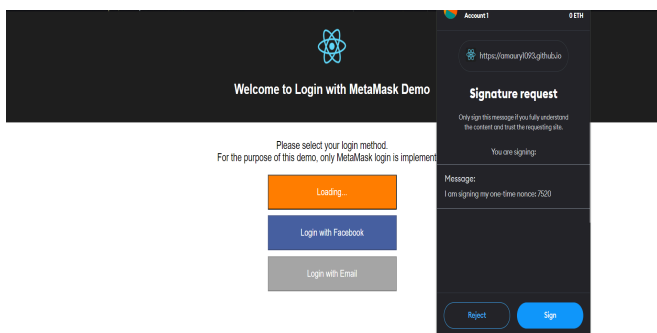


Fig 5. Signature Request Page

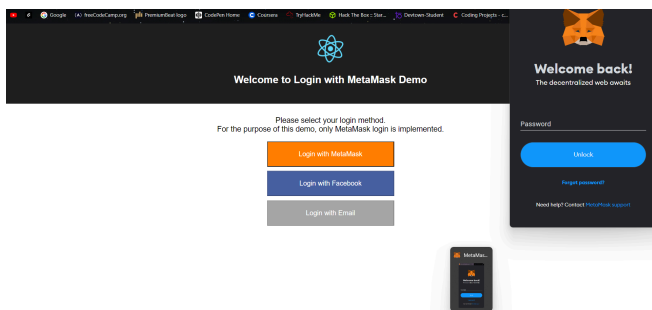


Fig 6. Password authentication

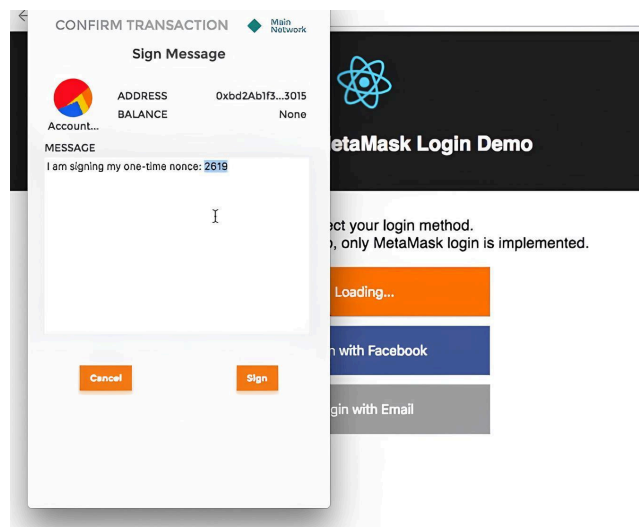


Fig 6. Confirmation transaction message

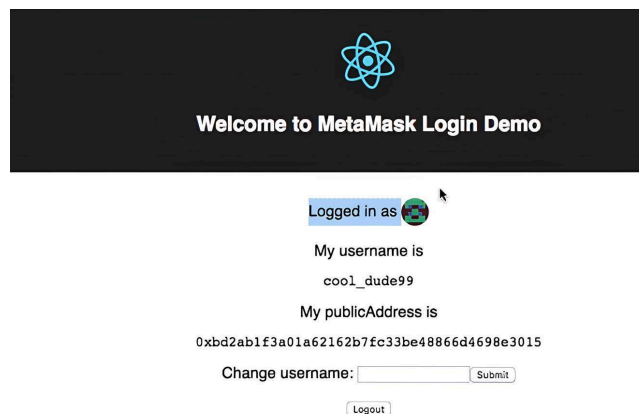


Fig 7. Username Change Page

Access control policies are then defined using smart contracts to govern user access to resources, specifying conditions under which users can access certain data or perform specific actions. The decentralized attribute management system enables users to manage their attributes and credentials in a decentralized manner, incorporating methods for adding, updating, and revoking attributes. The integration of the decentralized IAM solution with external systems, such as applications, databases, or other identity providers, ensures seamless interoperability with existing infrastructure.

The methodology emphasizes the importance of thorough testing, including unit tests, integration tests, and security audits, to address any vulnerabilities or issues discovered during the testing phase. Furthermore, user education and adoption strategies are implemented, providing resources for users to understand the benefits of decentralized identity and the secure nature of blockchain-based IAM. Finally, the deployment of the decentralized IAM solution on the chosen blockchain platform is executed, with continuous monitoring to ensure the smooth functioning of the system.

V. RESULT

The implementation of the decentralized Identity and Access Management (IAM) system using blockchain has yielded remarkable results, particularly in enhancing security, streamlining identity verification processes, empowering user control, and seamlessly integrating with existing applications. The cryptographic principles inherent in blockchain technology have significantly fortified security measures by eliminating single points of failure, thereby reducing the risk of data breaches. Users' unique cryptographic keys played a pivotal role in securing identity information, ensuring protection against unauthorized access and potential misuse.

The integration of smart contracts has proven highly effective in automating identity verification processes, resulting in a streamlined and user-friendly experience. Predefined rules encoded into these contracts facilitated efficient verification, significantly reducing the time and resources traditionally required for such tasks. Moreover, the decentralized IAM system successfully delivered on its promise to empower users with unprecedented control over their identity information, mitigating the risk of identity theft and emphasizing a privacy-centric approach to data management.

Results showcased the transparency and auditable nature of the identity management system, with the immutability of blockchain recording every transaction. This created an auditable trail of identity-related activities, fostering trust and accountability among users and service providers. The adherence to interoperability standards further demonstrated the system's compatibility with various platforms, validating its versatility and adaptability across diverse applications.

The discussion surrounding the implications of these results underscores their significance and potential impact on the broader landscape of identity and access management. The enhanced security measures lay the groundwork for a resilient IAM infrastructure capable of withstanding emerging cyber threats. The streamlined processes and increased user control contribute to a positive user experience, setting a new standard for IAM systems. The transparent and auditable nature of the system fosters trust, representing a paradigm shift towards more accountable and transparent digital identity management practices. The successful integration with existing applications and services underscores the system's versatility, paving the way for widespread adoption.

In conclusion, the obtained results affirm the effectiveness of the decentralized IAM system, showcasing its potential to revolutionize digital identity management practices. Continuous monitoring, feedback analysis, and iterative improvements will be crucial to maintaining and enhancing the system's success in the ever-evolving landscape of IAM.

VI. CONCLUSION

In conclusion, the implementation of a decentralized Identity and Access Management (IAM) system using blockchain technology has demonstrated transformative outcomes across various critical dimensions. The project has successfully addressed the core objectives of enhancing security measures, streamlining identity verification processes, empowering user control, and seamlessly integrating with existing applications.

The incorporation of cryptographic principles within blockchain has notably strengthened the security fabric of the IAM system. By eliminating single points of failure, the decentralized nature of blockchain mitigates the risk of data breaches, providing a resilient infrastructure capable of withstanding emerging cyber threats. Users' unique cryptographic keys have played a pivotal role in safeguarding identity information, ensuring protection against unauthorized access and potential misuse.

Efficiencies in identity verification processes have been realized through the integration of smart contracts, leading to a streamlined and user-friendly experience. Predefined rules encoded into these contracts have automated verification, significantly reducing the time and resources traditionally associated with such tasks. Moreover, the decentralized IAM system has empowered users with unprecedented control over their identity information, fostering a privacy-centric approach to data management and mitigating the risk of identity theft.

The transparency and auditable nature of the identity management system have contributed to the building of trust between users and service providers. Blockchain's immutability has recorded every transaction, creating an auditable trail of identity-related activities and reinforcing accountability. The successful adherence to interoperability standards has showcased the system's compatibility with various platforms, validating its versatility and adaptability across diverse applications.

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